

Novel modular approach based on Gaussian beam diffraction for analysing quasi-optical multi-reflector antennas

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Abstract: A novel approach for analysing quasi-optical multi-reflector systems is presented. In the proposed method of analysis the input beam (from the feed or output from a previous reflector) is expanded into a set of elementary Gaussian beams. These are propagated to the reflector. Reflection is treated in a geometrical optics manner. Diffraction is treated by applying the canonical problem of a 3-D Gaussian beam incident upon an opaque Kirchhoff half-screen. Its asymptotic solution yields a so-called boundary-diffraction wave that complements the geometrical wave that corresponds to the reflected beam. The output beam is a superposition of all reflected and diffracted elementary beams and can again be expanded into a set of Gaussian beams, making the method modular and suitable for large multi-element quasi-optical systems.

1 Introduction

In many antenna applications, e.g. radiometers and radio-astronomy applications [1, 2] several quasi-optical reflectors are cascaded to form a feed system for an electrically large main reflector. Such quasi-optical antenna systems are used at increasingly high frequencies up to the terahertz region. Methods for the design verification of quasi-optical (QO) systems are at present very limited. Geometrical optics (GO) does not account for the effects of edge diffraction. The geometrical theory of diffraction (GTD) is an extension to GO which includes diffraction effects but fails at caustics; this can be overcome by use of the uniform theory of diffraction (UTD). However, the method cannot easily be applied to multi-reflector systems because of its non-modularity due to multiple diffraction and caustics. In the context of this work modularity requires that the fields at the interface between two modules need to be fully specified. A module is defined as being an input field defined on an input surface (either from a feed or the output field from a previous module), a reflector and the subsequent output field defined on an output surface. Simply specifying the transverse fields on the output surface (preferably normal to the main collimated field) does not give the direction of any scattered ray that forms part of this field. The subsequent input field to the next module is thus incorrectly defined. The only way to fully specify the fields on the output surface, and hence replace the 'sources' inside that surface, is to invoke the field equivalence principle and specify currents on the surface. Integration of these currents is then required to find the field (or currents) on the next reflector. This is equivalent conceptually to physical optics (PO). It is for this reason that commercial multi-reflector antenna analysis software do not use UTD

for systems with more than two reflectors and then in a non-modular way.

Physical optics (PO) coupled with the physical theory of diffraction (PTD) is an accurate and modular method for analysing reflector antennas but is very expensive in terms of computation time and storage for electrically large multi-reflector systems. For the case of the near-field paraxial region of interest here we have found negligible difference between using PO and PO+PTD (undertaken using the GRASP8 software). Gaussian-beam optics has been successfully applied to the design of complete QO systems but it is unsuitable for full electromagnetic verification as it does not account for beam distortion and assumes negligible field at all reflector edges. Because of this most QO use excessively oversized reflectors to ensure a successful design. However, there is a growing need, especially in space-based systems, to make QO-fed systems as compact as possible.

Chou *et al.* [3, 4] have already responded to this by developing a PO-based Gaussian beam analysis suited to the rapid computation of the total (reflected plus diffracted) scattered field from an electrically large reflector. As in this work, it advances previous Gaussian beam methods of analysis by accounting for the presence of the rim of a reflector. Like Chou *et al.*, the solution here to the problem of rapid analysis of large reflectors is based upon reformulation of the illuminating source field in terms of a series expansion of Gaussian basis functions. However, the construction and application (and therefore physical interpretation of the operation) of the Gaussian beam expansion of the source field is entirely different.

We emphasise the key feature of modularity, to be elaborated upon below, as the significant development to the work of Chou *et al.* That is, the application of diffracted Gaussian beam analysis reported here, may be successively applied from reflector to reflector in a multi-antenna system. This prepares for future analysis of multi-antenna QO systems that may additionally incorporate intermediate signal processing components such as dichroics and wire grids.

A technical advantage is also present here in the mechanics of the analysis of the source field into Gaussian

beams. Chou *et al.* [4] acknowledge the advantage of a Gabor-type expansion (as an alternative to the method they used), to provide *a priori* a complete analysis of the source field and to remove any arbitrariness in the choice of initial beam parameters, as they have, once the expansion lattice over the plane of the source has been constructed. We avoid their concern of a Gabor-type analysis restricting control on the choice of initial beam parameters by using a windowed Fourier transform. This achieves a regular mesh in phase space, as they desire for their method. A wavelet basis would, however, generate the irregular phase space mesh that they seek to avoid, for convenience of engineering application.

Further comment on the additional matters related to the effects of double curvature in a reflector, upon reflection and diffraction processes, are respectively held over until Sections 2.2 and 2.3.

The diffracted Gaussian beam approach (DGBA) tries to combine the numerical efficiency of a description in terms of beams, similar to the ray description used in GO, with the modularity and reliability of physical optics. In fact the DGBA may be regarded as a UTD-like operation, but now with PO-based Gaussian beams rather than with the conventional rays of GO, thereby being continuously operative over the entire face of the reflector from regions of focused (or nearly focused) fields across to the edge.

A single step in the DGBA multi-reflector analysis is illustrated in Fig. 1. The incident field on an input plane, which could be either the near field of a feed or the radiated field from a previous QO reflector, is expanded in terms of fundamental 3-D Gaussian beam modes, each beam having a beamwidth of several wavelengths. The elementary beams are propagated to the reflector to be analysed. The change of the complex beam parameters by the reflector's curvature is calculated according to the laws of geometrical optics. The reflected beams are then propagated to the output plane. Those beams that hit the reflector at less than two beamwidths distance from the rim, suffer significant diffraction. Diffraction is treated by considering the edge at the nearest rim point to be straight. The asymptotic solution of the canonical problem of a Gaussian beam incident upon an opaque Kirchhoff half-screen [5] provides a diffracted field in terms of complementary error functions which complements the GO reflected beam. This is analogous to the GTD approach where a locally plane

wave is incident. The combined reflected and diffracted field terms give a regular and continuous field within the paraxial region of the reflected beam. Finally, all reflected and diffracted beams are superposed on an output plane. The total field on the output plane is again cast into a Gaussian beam expansion, making the method highly modular.

2 Theory

Here, the individual steps of a single reflector analysis with the diffracted Gaussian beam approach will be described in detail. These are: the expansion in terms of Gaussian beams, the reflection of Gaussian beams at curved surfaces, and the diffraction of Gaussian beams.

2.1 Gaussian beam expansion

In [6] a very detailed description of the properties of the so-called windowed Fourier transform that was originally developed for signal analysis is given. This consists in multiplying the spatially shifted signal with a usually compactly supported window function and computing the Fourier coefficients of the product. The windowed Fourier transform as presented in [6] offers a higher degree of freedom than the well-known Gabor expansion [7]. Unlike Gabor's series representation, the spatial sampling rate can be chosen higher than according to the Nyquist criterion, and this oversampling leads to an advantage in terms of stability and convergence [8]. In [6] an algorithm to calculate the so-called 'dual frame function' for a 1-D Gaussian shaped window function is derived. The Gaussian beam expansion coefficients are then readily obtained by solving a convolution integral between the 1-D complex-valued signal and the real-valued dual-frame function. In our application, we have the two transverse components E_x , E_y , of the electric field given on a plane $z=0$ and want to expand the field in terms of circular beams. Generalising the expansion in [6] to a 2-D signal results in an expansion of the form

$$f(x, y) = \sum_m \sum_n \sum_\mu \sum_\nu A_{m\mu\nu} w_{2-D}(x - mL_0, y - \mu L_0) \exp(j(n\Omega_0 x + \nu\Omega_0 y)) \quad (1)$$

with a Gaussian shaped window function

$$w_{2-D}(x, y) = \frac{\sqrt{2}}{L} \exp(-\pi(x^2 + y^2)/L^2) = w(x)w(y) \quad (2)$$

where f stands for either of the two transverse field components, L is the radius of the circular Gaussian beam and $A_{m\mu\nu}$ is the expansion coefficient. The product $\Omega_0 L_0$, where L_0 represents the spatial shift and Ω_0 represents the spectral shift, has to be chosen smaller than 2π (Gabor's representation) to ensure stability [8]. The smaller it is compared to 2π , the smoother the shape of the dual frame function, it then resembling more and more the Gaussian shape of the window function. Since the window function can be separated into the two transverse co-ordinates, so the 2-D dual frame function $\tilde{w}_{2-D}(x, y)$ can also be written as the product of two 1-D dual frame functions. The expansion coefficients are then calculated as

$$A_{m\mu\nu} = \iint f(x, y) \tilde{w}(x - mL_0) \tilde{w}(y - \mu L_0) \times \exp(-j(n\Omega_0 x + \nu\Omega_0 y)) dx dy \quad (3)$$

where $\tilde{w}$ is the dual frame function. Since $\tilde{w}$ can be computed by evaluating a series expansion, determination of the expansion coefficients is best implemented by using the numerical efficiency of the two-dimensional FFT. Let the expansion be characterised by N contributing spectral

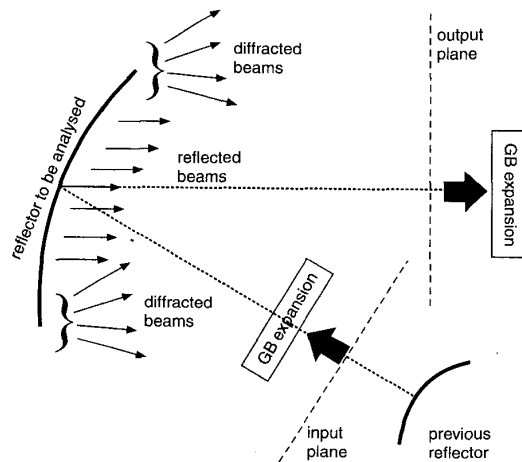


Fig. 1 Illustration of a single step of the DGBA in a multi-reflector analysis

terms and M contributing spatial terms per dimension such that $|n|, |v| \leq N/2$ and $|m|, |\mu| \leq M/2$. Let $\Lambda = 2\pi/\Omega_0$ be the spatial period corresponding to the spectral frequency Ω_0 . A numerical evaluation (FFT) of the Fourier integral (see (3)) is based on the following identity:

$$\bar{A}_{mn\mu\nu} = \frac{\Lambda^2}{N^2} \sum_{i=0}^{N-1} \sum_{k=0}^{N-1} \bar{f}_{mn\mu\nu} \exp(-j2\pi(in + kv)/N) \quad (4a)$$

where

$$\bar{f}_{mn\mu\nu} = \sum_i \sum_k f((i + n/N)\Lambda, (k + v/N)\Lambda) \tilde{w}((i + n/N)\Lambda - mL_0) \tilde{w}((k + v/N)\Lambda - \mu L_0) \quad (4b)$$

From the samples $\bar{A}_{mn\mu\nu}$ returned by the FFT algorithm the true expansion coefficients

$A_{mn\mu\nu}$ are simply recovered by a simple re-indexing:

$$A_{mn\mu\nu} = \begin{cases} A_{mn\mu\nu} & 0 \leq n \leq N/2; \quad 0 \leq v \leq N/2 \\ A_{mN+n\mu\nu} & -N/2 \leq n < 0; \quad 0 \leq v \leq N/2 \\ A_{mn\mu\nu+N} & 0 \leq n \leq N/2; \quad -N/2 \leq v < 0 \\ A_{mn+N\mu\nu+N} & -N/2 \leq n < 0; \quad -N/2 \leq v < 0 \end{cases} \quad (5)$$

Based on the algorithm above, a block diagram realisation of the two-dimensional FFT processor of the expansion coefficients is depicted in Fig. 2. The quantity Q corresponds to the number of summands (ik) in (4b) which depends largely on the smoothness of the dual frame function as dictated by the product $\Omega_0 L_0$, the smoother the function, the smaller i and k .

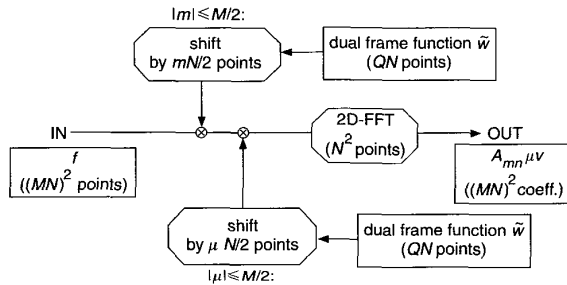


Fig. 2 2-D FFT implementation of the Gaussian beam expansion processor

To describe the scalar field components E_x , E_y of each contributing term in the expansion everywhere in the half space $z \geq 0$, asymptotic techniques are employed. For the Gaussian shape of the window function, they yield simple and, under most circumstances, accurate results, which are readily interpretable in terms of complex beams and are paraxially equivalent to a complex source point response. Let $B_{mn\mu\nu}(x, y, z)$ denote the beam mode corresponding to the expansion coefficient $A_{mn\mu\nu}$ in (1) such that

$$f(x, y, z) = \sum_{m,n,\mu,\nu} [A_{mn\mu\nu} \exp(j(mn + \mu\nu)\Omega_0 L_0)] B_{mn\mu\nu}(x, y, z); \quad z \geq 0 \quad (6)$$

Asymptotically we obtain

$$B_{mn\mu\nu}(\rho_t, z_t) \cong \frac{\sqrt{2}}{L} \frac{q_{nv}(0)}{q_{nv}(z_t)} \exp\left(jk_0\left(z_t + \frac{\rho_t^2}{2q_{nv}(z_t)}\right)\right) \quad (7)$$

The transformed coordinates ρ_t and z_t are linked with the Cartesian co-ordinates x, y, z and the spectral components

of the wavevector via the transformation

$$z_t = [n\Omega_0(x - mL_0) + v\Omega_0(y - \mu L_0) + \sqrt{k_0^2 - (n\Omega_0)^2 - (v\Omega_0)^2}z]/k_0; \quad k_0 = 2\pi/\lambda \quad (8a)$$

$$\rho_t^2 = (x - mL_0)^2 + (y - \mu L_0)^2 + z^2 - z_t^2 \quad (8b)$$

z_t can be viewed as the projection of the vector pointing from the source point $(mL_0, \mu L_0, 0)'$ to the observation point $(x, y, z)'$ onto the direction

$$\hat{z}_t = \frac{1}{k_0} \left(n\Omega_0, v\Omega_0, \sqrt{k_0^2 - (n\Omega_0)^2 - (v\Omega_0)^2} \right)'$$

of the wavevector. The complex beam parameter $q_{nv}(z_t)$ absorbs all the information about the local spot size $\omega(z_t)$ and the local radius of curvature $R(z_t)$ of the beam at each axial position z_t :

$$q_{nv}(z_t) = \left(\frac{1}{R(z_t)} + j \frac{\lambda}{\pi \omega^2(z_t)} \right)^{-1} = z_t - j \frac{L^2}{\lambda} \left(1 - \left(\frac{nK_0}{k_0} \right)^2 - \left(\frac{vK_0}{k_0} \right)^2 \right) \quad (9)$$

It is noted that the asymptotic expression (7) in terms of a complex beam is accurate only within the range of the paraxial approximation. Fig. 3 illustrates this interpretation of the expansion in terms of Gaussian beam modes, as a number of spectral Gaussian beam modes located at each spatial 'sampling point' separated by L_0 .

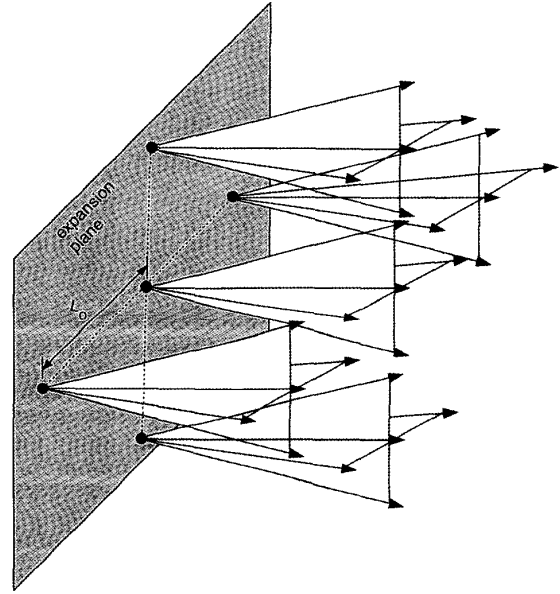


Fig. 3 Interpretation of the expansion in terms of Gaussian beam modes

2.2 Gaussian beam reflection

Geometrical optics is applied to determine how the complex beam parameter q changes when a Gaussian beam impinges upon a curved reflector surface. At the reflection point, the spot size of the beam stays the same while the curvature, which can generally be described by a set of two principal radii of curvature R_1 , R_2 changes according to the local curvature of the reflecting surface.

We consider a point $\vec{r}_0$ on a surface, which could be either a reflector surface or a wavefront, and locate a right-

handed co-ordinate system $(\hat{\tau}_1, \hat{\tau}_2, \hat{n})$ at the point, with $\hat{n}$ being the surface normal. $\hat{\tau}_1$ and $\hat{\tau}_2$ denote the two principal directions of curvature of the surface, and it is clear that one can describe the surface locally around the point $\vec{r}_0$ by the biparabolic expansion

$$\vec{r}(u, v) = \vec{r}_0 + u\hat{\tau}_1 + v\hat{\tau}_2 - \frac{1}{2} \left(\frac{u^2}{R_1} + \frac{v^2}{R_2} \right) \hat{n} \quad (10)$$

where R_1 and R_2 are the two principal radii of curvature. The two principle directions $\hat{\tau}_1$ and $\hat{\tau}_2$ are uniquely described by one angle ψ , which denotes the rotation of $\hat{\tau}_1$ in the plane normal to $\hat{n}$ with respect to a reference vector $(\hat{y} \times \hat{n})/|\hat{y} \times \hat{n}|$ in a mathematically positive sense. Hence for all local points of a wavefront or of a reflector surface the local curvature can be characterised by a set of curvature parameters $(\hat{n}, R_1, R_2, \psi)$.

Let us consider the case of an incident differential ray strip with local curvature parameters $(\hat{s}^i, R_1^i, R_2^i, \Psi^i)$ that impinges upon a reflector with local curvature parameters $(\hat{n}, R_1^c, R_2^c, \Psi^c)$ at an incident angle θ relative to the surface normal (Fig. 4). We want to determine the curvature parameters $(\hat{s}^r, R_1^r, R_2^r, \Psi^r)$ of the reflected ray strip. The direction of the reflected ray, using Snell's law, is

$$\hat{s}^r = \hat{s}^i + 2 \cos(\theta) \hat{n} \quad (11)$$

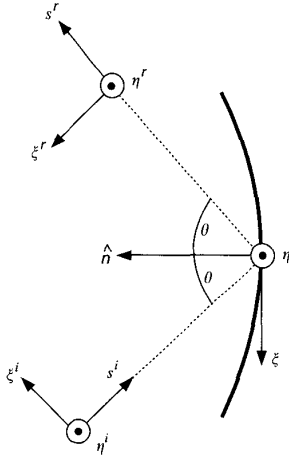


Fig. 4 Reflection of a Gaussian beam at a curved surface: geometry

It is convenient to do all calculations in the ray-based co-ordinate system $(\xi^{i,c,r}, \eta^{i,c,r})$ as illustrated in Fig. 4. $(u^{i,c,r}, v^{i,c,r})$ from (10) are simply related to $(\xi^{i,c,r}, \eta^{i,c,r})$ by the transformations

$$\begin{pmatrix} u^{i,c,r} \\ v^{i,c,r} \end{pmatrix} = \begin{bmatrix} \cos(\alpha_{i,c,r}) & \sin(\alpha_{i,c,r}) \\ -\sin(\alpha_{i,c,r}) & \cos(\alpha_{i,c,r}) \end{bmatrix} \begin{pmatrix} \xi^{i,c,r} \\ \eta^{i,c,r} \end{pmatrix} \quad (12)$$

The rotation angles $\alpha_{i,c}$ follow from simple geometrical calculations since $\Psi^{i,c}$ and the unit vectors of the ray-based co-ordinate system are known. Matching the phase between the incident and the reflected wavefront on the reflector surface [9] and dropping all third and higher order terms results in an equation system for the curvature parameters

R_1^r, R_2^r, α_r of the reflected ray strip:

$$\begin{pmatrix} \Sigma_r \\ \cos(2\alpha_r) \Delta_r \\ \sin(2\alpha_r) \Delta_r \end{pmatrix} = \begin{pmatrix} \Sigma_r \\ \cos(2\alpha_i) \Delta_i \\ -\sin(2\alpha_i) \Delta_i \end{pmatrix} + \begin{bmatrix} 1/\cos \theta + \cos \theta & 1/\cos \theta - \cos \theta & 0 \\ 1/\cos \theta - \cos \theta & 1/\cos \theta + \cos \theta & 0 \\ 0 & 0 & 2 \end{bmatrix} \times \begin{pmatrix} \Sigma_c \\ \cos(2\alpha_c) \Delta_c \\ \sin(2\alpha_c) \Delta_c \end{pmatrix}$$

with

$$\begin{aligned} \Sigma_{i,c,r} &= (1/R_2^{i,c,r} + 1/R_1^{i,c,r})/2 \text{ and } \Delta_{i,c,r} \\ &= (1/R_2^{i,c,r} - 1/R_1^{i,c,r})/2 \end{aligned} \quad (13)$$

Ψ^r is easily calculated from α_r , by computing the principal directions of the reflected ray strip in terms of the ray-based co-ordinate system and taking the scalar product with the reference vector. The reflected beam is approximated by a circular beam with a complex beam parameter

$$q = \left(\frac{1}{\sqrt{R_1^r R_2^r}} + j \frac{\lambda}{\pi w^2} \right)^{-1} \quad (14)$$

and the diffraction theory for circular Gaussian beams, as presented in the next subsection, is applied.

Applying the conventions of this subsection, the numerical results in Section 3 for the example of an offset Cassegrain configuration have met criteria that establish a compromise between the local spot size of an elementary beam being small with respect to the radii of curvature of the reflector, and being large enough to maintain a paraxial beam condition as it propagates to the reflector. Specifically, a support width (or window) for the Gaussian kernel is chosen in proportion to the reflector diameter (D), and displacement of the reflector from the source aperture, such that a beam spot size of 6.4λ (at 30 GHz) results when the field is sampled at the Nyquist rate $\lambda/2$. The far field distance $2D^2/\lambda$, where beam spreading begins, is 0.82 m. This determines, for example, the location of the input expansion plane that illuminates the first reflector.

2.3 Gaussian beam diffraction

In [5], diffraction of a Gaussian beam with a circular spot size normally incident upon an opaque Kirchhoff half-screen was investigated based on the boundary-diffraction wave (BDW) theory. There, the evaluation of the boundary-diffraction wave by the steepest descent method yielded the uniform asymptotic representation of the total diffracted field consisting of the geometrical optics component (geometrical wave) and diffraction components in terms of complementary error functions (boundary-diffraction wave). An attempt was made to extend the analysis of [5] to elliptical Gaussian beams. Unfortunately, an asymptotic representation of the Kirchhoff integral is prohibited, the poles being anti-symmetric with respect to the stationary point. We note, however, that proper account of the effect of reflector bi-curvature upon the BDW may be made by otherwise expressing the amplitude and phase components of the field quantity of the Kirchhoff integral in terms of a curvature (or phase) matrix that has complex rather than real-valued elements [3]. In this respect too, (13) may be modified to describe general reflection from a bicurvature reflector by being derived from a curvature matrix having

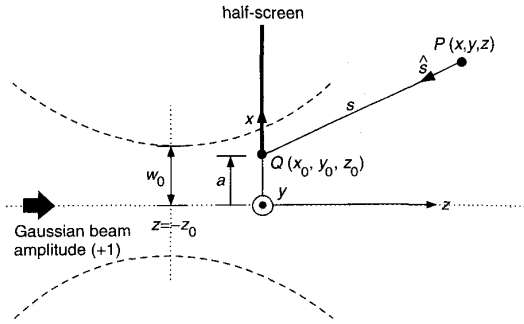


Fig. 5 Geometry of the canonical problem solved in [5] (forward-scattering region)

complex rather than real-valued elements. Fig. 5 depicts the geometry of the problem. The incident beam is assumed to propagate in the z -direction. $P(x, y, z)$ is the observation point and $Q(x_0, y_0, z_0)$ is a source point of boundary-diffraction located on the boundary Γ of the half screen.

2.3.1 Field representation in the backward-scattering region: The solution of this canonical problem is not repeated since it can be found in [5]. Instead, we show how the solution can be adopted in the context of a single reflector analysis. The formulation presented in [5] cannot be used directly since we are interested only in the diffracted field in the backward-scattering region ($z < 0$). We try to find an equivalent geometry to describe the diffracted field in the backward-scattering region. Firstly, in addition to the incident wave we have a reflected wave in the backward-scattering region. This can be characterised by an image beam at the opposite side of the half-screen, the same in magnitude but opposite in sign to the incident beam. Secondly, the reflected beam is propagating in the $(-z)$ -direction and diffraction has to be described in a new co-ordinate system with the transformed x -, y - and z -axes pointing in the $(-x)$ -, y - and $(-z)$ -directions, respectively. Thirdly, in the equivalent geometry the equivalent half-screen must be complementary to the true half-screen to generate a shadow region in the lower half-space ($x < 0$). The boundary diffraction wave, however, does depend only on the location of the edge and not on the orientation of the half-screen. Fig. 6 shows the equivalent geometry for calculating the diffracted field for the geometry shown in Fig. 5 in the backward-scattering region.

To check the accuracy of the asymptotic method, the problem of a normally incident Gaussian beam upon a

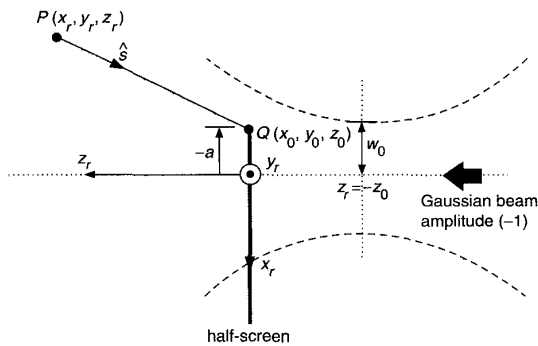


Fig. 6 Equivalent geometry for determining the diffracted field in the backward-scattering region

Kirchhoff half-screen has been analysed with the commercial reflector analysis tool GRASP8 [10], using physical optics. Fig. 7 shows the field in the backward-scattering region for different beamwidths in comparison with GRASP. The fields are calculated for a near field cut with the co-ordinates ($y=0$; $z=-300\lambda$). The parameter a in Fig. 5 is chosen to be zero. It is seen from the results that the boundary diffraction theory is very accurate near the beam axis. Outside the paraxial region, the results start to deviate from the physical optics results but even there the deviation is not more than a decibel or two.

Fig. 8 shows a comparison with GRASP physical optics for various values of the parameter a (see Fig. 5). The field has been calculated at the same near field cut as in Fig. 7 and the half-beamwidth w_0 has been chosen to be eight wavelengths. Again, the agreement within the paraxial region is excellent and outside the paraxial region the agreement is still reasonably good, keeping in mind that the field levels are low.

2.3.2 Generalisation to oblique incidence: Assume that the direction of propagation of the incident beam is still perpendicular to the edge of the half-screen. However, the orientation of the half-screen may now be non-perpendicular to the direction of propagation. Fig. 9 depicts the geometry of the discussed problem, showing all involved co-ordinate systems. We have three different co-ordinate systems: the transmitted beam-related co-ordinate system (x_t, y_t, z_t), the reflected beam related co-ordinate system (x_r, y_r, z_r) and the half-screen based co-ordinate system (x, y, z). The origins of the transmitted and reflected beam related co-ordinate systems are such that the location of the edge corresponds to a zero axial position in both co-ordinate systems. The origin of the half-screen based co-ordinate system is the intersection point between the incident and reflected beam axes and the plane of the half-screen. Since normal incidence is assumed with respect to the polar angle, the y -direction is the same for all three co-ordinate systems. Let φ be the incidence angle, being defined as the angle between the negative of the direction of propagation ($-\hat{z}_t$) and the surface tangential unit vector normal to the edge of the half-screen ($\hat{z}$). The observation point is described in the half-screen based cylindrical (ρ, φ, y)-coordinate system. The transmitted and reflected beams propagate in the $(\pi + \varphi_0)$ - and $(\pi - \varphi_0)$ -direction, respectively. Their co-ordinate systems are expressed in terms of the half-screen co-ordinate system by simple translation and rotation around the y -axis:

$$\begin{pmatrix} z_r \\ x_r \end{pmatrix} = \rho \begin{pmatrix} \cos(\varphi - (\pi + \varphi_0)) \\ \sin(\varphi - (\pi + \varphi_0)) \end{pmatrix} + \begin{pmatrix} z_e \cos(\varphi) \\ 0 \end{pmatrix} \quad (15a)$$

$$\begin{pmatrix} z_r \\ x_r \end{pmatrix} = \rho \begin{pmatrix} \cos(\varphi - (\pi - \varphi_0)) \\ \sin(\varphi - (\pi - \varphi_0)) \end{pmatrix} + \begin{pmatrix} z_e \cos(\varphi_0) \\ 0 \end{pmatrix} \quad (15b)$$

Since the boundary diffraction wave is obtained as a line integral along the edge and does not depend on the orientation of the half-screen, we are now in a position to apply the equations governing the original geometry in Fig. 5 in terms of the reflected beam related co-ordinate system (x_r, y_r, z_r). The offset distance a from the edge (see Fig. 5) is related to the axial position z_e of the edge in the half-screen based co-ordinate system (Fig. 9) by $a = z_e \sin(\varphi_0)$.

Fig. 10 shows a comparison of the theory with GRASP physical optics for various orientations of the half-screen as

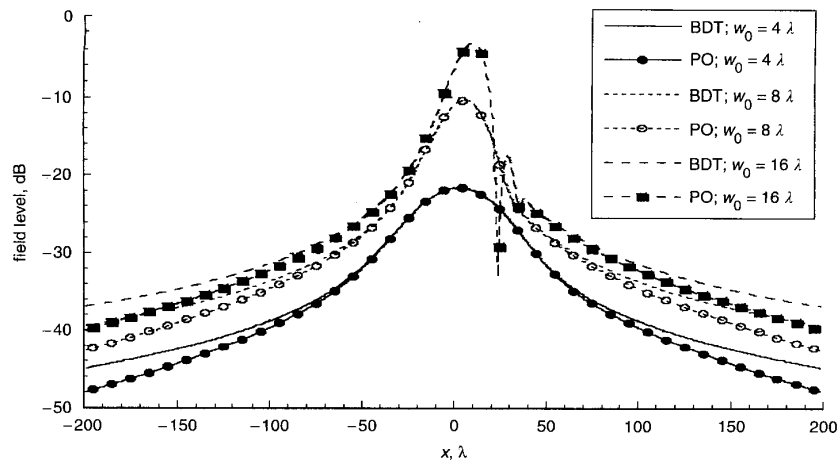


Fig. 7 Comparison of the backscattered field with physical optics for various beam spot sizes
 $a = 0$, $z = -300\lambda$

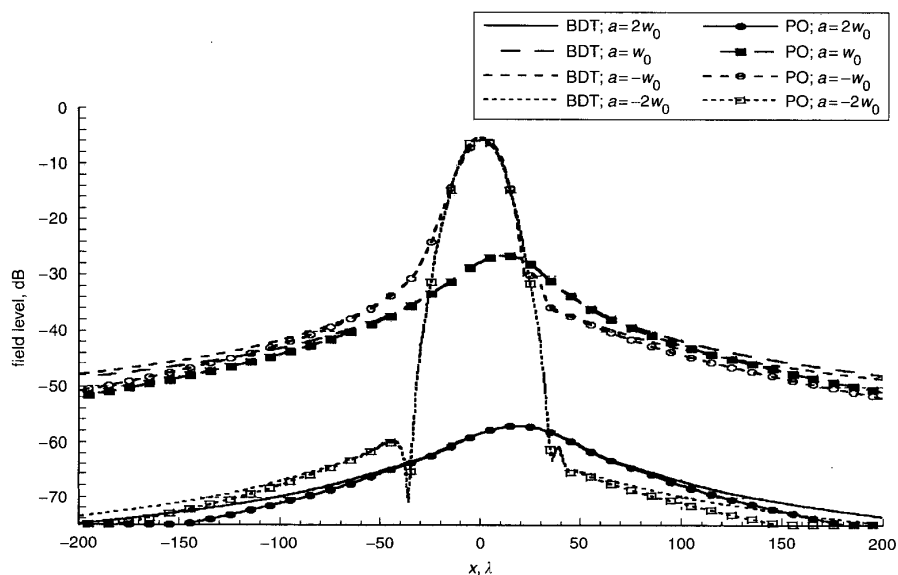


Fig. 8 Comparison of the backscattered field with physical optics for various values of the parameter a (for geometry in Fig. 5)
 $w_c = 8\lambda$, $z = -300\lambda$

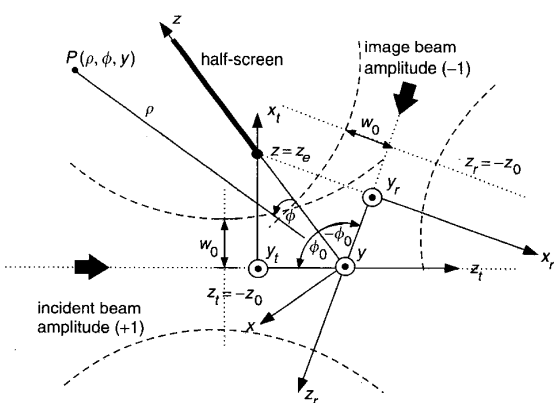


Fig. 9 Equivalent geometry for determining the diffracted field in the backward-scattering region at oblique incidence

being characterised by the angle ϕ_0 . With respect to the polar angle, normal incidence is assumed ($\theta = 90^\circ$). The field has been calculated at the same near field cut as in Fig. 7 and the half-beamwidth w_0 has been chosen to be eight wavelengths. The parameter a equals zero. From the results it is seen that the agreement within the paraxial approximation is still very good, although there are some slight deviations even near the reflected beam centre for non-normal incidence. They grow progressively for more oblique angles.

It would be desirable to derive a solution for non-normal incidence with respect to the polar angle (the angle θ in the yz plane of Fig. 9). In the classical GTD canonical problem of a plane wave incident on a half-screen the field components along the edge can be separated from the transverse components allowing the solution for normal incidence to be used to construct a general solution. Unfortunately, in our general problem the incident

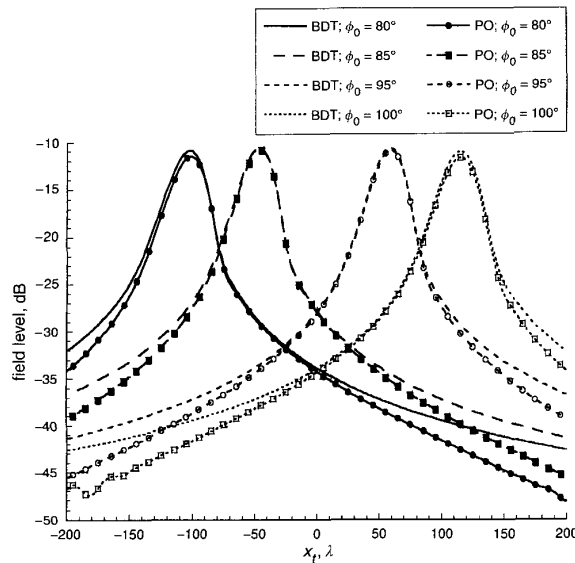


Fig. 10 Comparison of the backscattered field with physical optics for various orientations of the half-screen
 $w_0 = 8\lambda$, $\alpha = 0$, $z_t = -300\lambda$

Gaussian beam cannot be separated into three Cartesian co-ordinates. It can, however, be shown that for non-normal incidence with respect to the polar angle within a first-order approximation the same stationary point and the same shadow boundary and the same poles are obtained as for normal incidence. Our conclusion is that for incidence angles close to 90° the results for the equivalent geometry in Fig. 9 can still be applied. For θ incidence, the beam direction of the transmitted and reflected beams change according to geometrical optics and the transmitted and reflected beam related co-ordinate systems are now rotated by $(\pi/2 - \theta)$ around the x -axis compared to the case with normal incidence. Geometrically, the geometrical optics beams are rotated together with their associated diffraction components. Of course, this treatment involves an approximation error, which is very small for almost normal incidence and gets larger for more oblique incidence angles. It is noted, that in quasi-optical systems the distance between the reflectors is usually much larger than their diameter such that the beams are well within the validity of the paraxial approximation and hence the condition of near normal incidence is well fulfilled.

3 Numerical Results

To validate the proposed method and to show its modularity the offset Cassegrain configuration shown in Fig. 11 was analysed at 30 GHz. Field plots on various near field planes are shown and the results are compared to those of a physical optics analysis with the commercial reflector analysis package GRASP. At 30 GHz the subreflector and main reflector have diameters of approximately 40λ and 100λ , respectively.

The far field pattern of the feed was assumed to be Gaussian illuminating the subreflector edge at a level of -6 dB. The structure was analysed in two steps. The input field plane for the analysis of the subreflector was chosen 0.8 m away from the feed. The elementary beams in the Gaussian beam expansion were eight wavelengths wide.

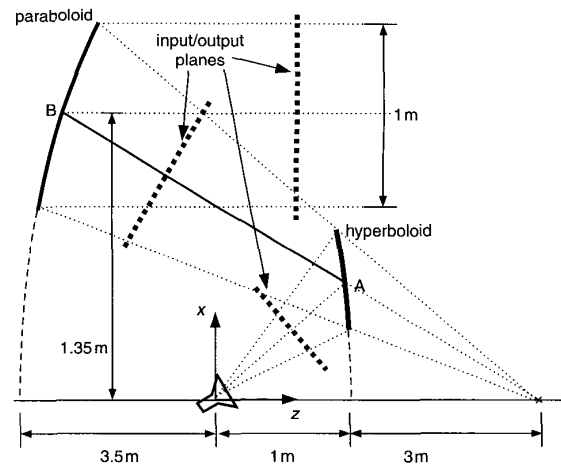


Fig. 11 Geometry of a Cassegrain antenna test case

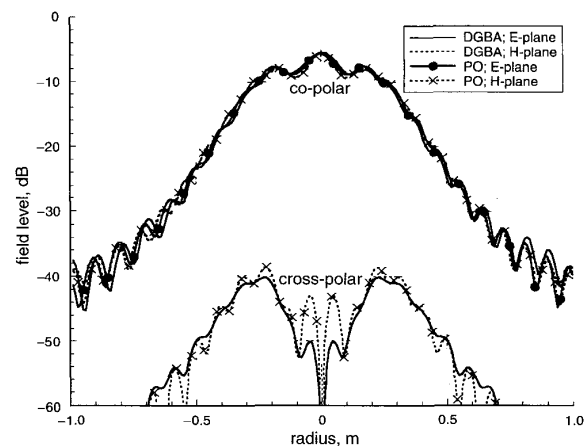


Fig. 12 Comparison of the near field from the subreflector at 30 GHz with the results of physical optics

The output plane was chosen to be 2 m away from the subreflector in a plane perpendicular to the line AB. Fig. 12 shows the output field in the offset plane in comparison to PO calculations obtained with the commercial reflector analysis code GRASP. The agreement is excellent. In the second step the output field of the subreflector was expanded into a new set of Gaussian beams. These were traced to the main reflector employing the diffracted Gaussian beam analysis. Fig. 13 shows the electric field in the offset plane at a distance of 4.5 m from the main reflector. Again, the agreement with PO results is very good.

4 Conclusions

A novel diffracted Gaussian beam approach for analysing a quasi-optical multi-reflector system has been proposed. The method is especially well suited to the analysis of electrically large multi-reflector systems where physical optics is computationally expensive. As opposed to GTD, the new method is highly modular and does not suffer from caustic problems. It has been tested for an offset Cassegrain configuration and good agreement with PO has been obtained.

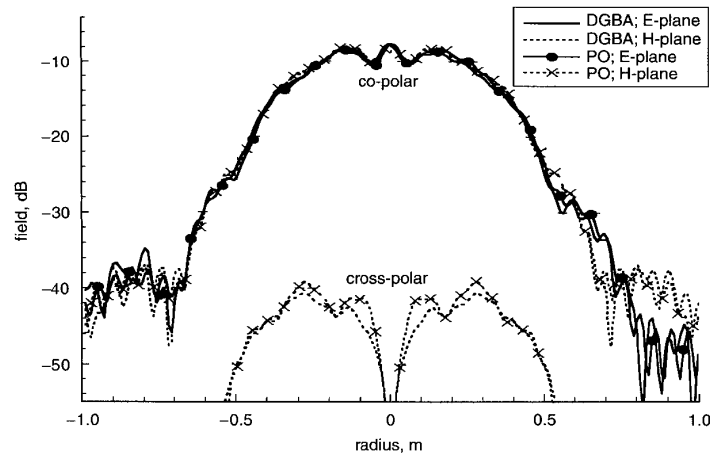


Fig. 13 Comparison of the near field from the main reflector at 30 GHz with the results of physical optics

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